# LAB | Amazon DynamoDB Basics Extra

## Introduction

In this hands-on lab, you will learn how to use **Amazon DynamoDB**, a fully managed NoSQL database service designed for fast and scalable applications. Unlike traditional relational databases, DynamoDB uses a **key-value** and **document-based** storage model, making it highly flexible and performant.

## Objectives

By the end of this lab, you will:

- Understand the basics of **DynamoDB** and its key features.
- Create a **DynamoDB table** with a primary key and optional sort key.
- Insert, retrieve, and query data in DynamoDB.
- Explore indexing and how to filter data efficiently.

## Prerequisites

- **AWS Account**: Ensure you have access to an AWS account.
- **Basic Knowledge**: Familiarity with NoSQL concepts and AWS services.
- **AWS CLI (Optional)**: For command-line interactions, but not required.

## Step 1: Open the DynamoDB Console

1. **Sign in to AWS Management Console**.
2. Search for **DynamoDB** in the AWS search bar and open the service.

## Step 2: Create a New DynamoDB Table

1. Click on **Create Table**.
2. **Configure Table Details**:
   - **Table Name**: `OnlineStore`
   - **Partition Key** (Primary Key): `CustomerID` (Type: `String`)

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- **Sort Key (Optional)**: `OrderID` (Type: `Number`) - Helps organize orders for each customer.

3. **Capacity Mode**:
   - Select **On-Demand** (recommended for beginners to avoid managing throughput manually).
4. **Create the Table** and wait for it to be provisioned.

## Step 3: Insert Data into the Table

1. Navigate to the **Items** tab.
2. Click **Create Item** and enter the following sample data:

```
{
  "CustomerID": "C001",
  "OrderID": 101,
  "Name": "John Doe",
  "Email": "john@example.com"
}
```

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3. Click **Save Item**.
4. Repeat the process to add more data.

## Step 4: Query Data in DynamoDB

1. Navigate to the **Explore Table** tab.
2. Search for **CustomerID = C001** to retrieve John Doe's details.
3. If you added multiple orders per customer, experiment by filtering using **OrderID**.

## Step 5: Create a Secondary Index (Optional)

1. **Go to Indexes** in your DynamoDB table.
2. Click **Create Index** and configure:
   - **Partition Key**: `Email` (Type: `String`)
   - **Sort Key** (Optional): `OrderID` (Type: `Number`)
   - **Projection Type**: Include all attributes.
3. Click **Create Index** and wait for it to be provisioned.
4. Use the index to **query data by Email** instead of CustomerID.

## Summary

In this lab, you:

1. Created a **DynamoDB table** with a primary key (`CustomerID`).

2. Inserted **sample data** into the table.
3. Queried data using **partition keys and filters**.
4. (Optional) Created a **secondary index** to retrieve data using alternative keys.

DynamoDB is a powerful, scalable NoSQL database that simplifies data storage for applications requiring high-speed access. Unlike traditional databases, it is **fully managed**, eliminating the need for server maintenance.

## Next Steps

- Try **batch writing** multiple items at once using AWS SDKs.
- Implement **DynamoDB Streams** to track real-time changes.
- Explore **Global Tables** for multi-region replication.

🚀 Keep experimenting with DynamoDB to unlock its full potential!

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Extra

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